Remove Duplicates From Linked List

AlgoExpert · Easy · Linked Lists

The problem

- List is sorted; remove nodes with repeated values in place.

Intuition

- The list is sorted, so duplicates are adjacent; we can skip them in place.

Approach

- 1. Walk the list
- 2. While the next node has the same value, bypass it
- 3. Move to the next distinct node
- 4. Return the head

Java solution

```java
public class RemoveDuplicatesFromLinkedList {
    public static LinkedListNode removeDuplicatesFromLinkedList(LinkedListNode head) {
        LinkedListNode cur = head;
        while (cur != null) {
            LinkedListNode next = cur.next;
            while (next != null && next.value == cur.value) next = next.next;
            cur.next = next;
            cur = next;
        }
        return head;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- The list is sorted, so duplicates are adjacent; we can skip them in place.
- Complexity: O(n)